CIS3700

Assignment 2

PART A

Describe how the brute force BFS program operates:

Each board configuration is a state, it is stored in an immutable tuple so it can be tracked in a visited set.

A queue is used to store states along with the sequence of moves taken to reach them. At each step, it removes the oldest state from the queue, checks whether it is a goal state, and if not, generates all legal successor states. Any successor that has not been visited is added to the queue. This happens until the red car exits the board.

Because BFS explores states strictly in order of increasing depth and all moves have equal cost, the first solution found uses the fewest possible moves. It does not use any heuristic guidance.

Provide a convincing argument that each heuristic is admissible.

H1: The red car has to physically move one cell at a time towards the exit so the number of remaining columns between the red car and the exit is a lower bound on the number of moves required.
Since this heuristic ignores blocking vehicles, it can underestimate the true cost but it doesn't overestimate it.

H2: Adding the number of blocking vehicles to the distance estimate still underestimates or equals the actual cost, because each blocking vehicle occupying the exit row needs to be moved before the red car can pass and it doesn't account for additional moves that might be required to free the blocking vehicles.

Therefore both heuristics are admissible.

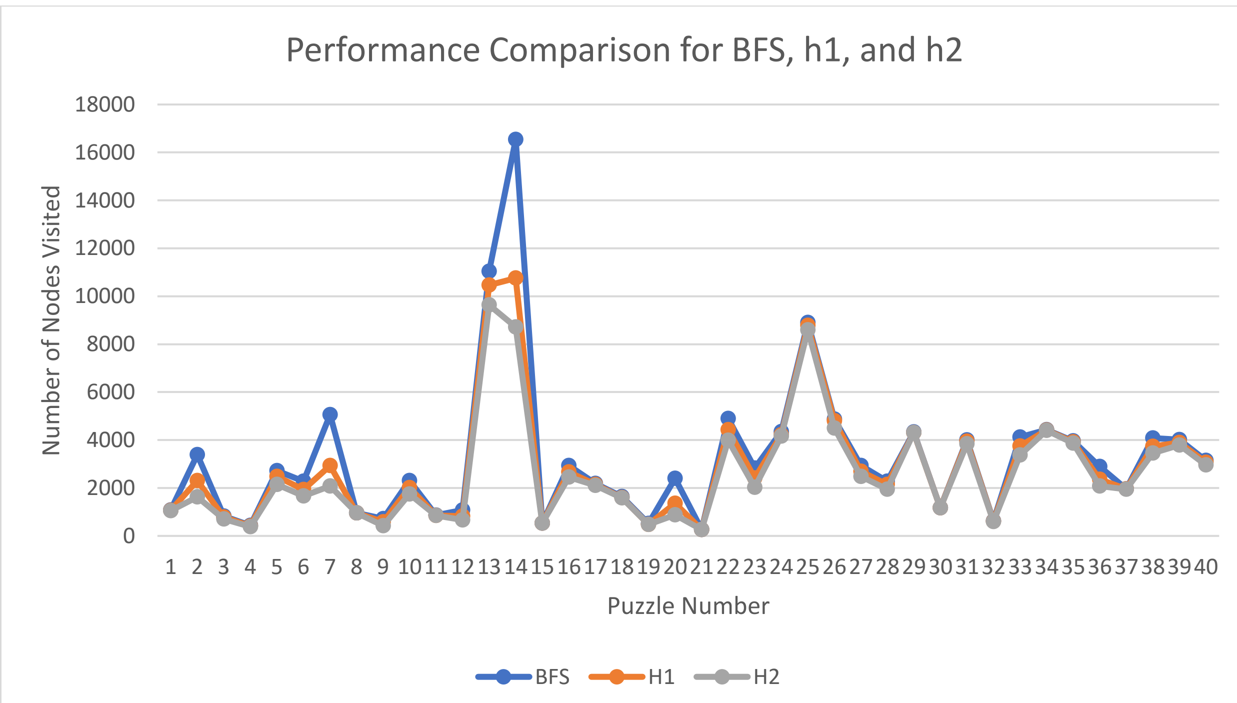
Provide a convincing argument that one of your heuristics gives consistently higher (or occasionally equally high) admissible estimates than the other.

According to the lecture principle of highest of the low, the best admissible heuristic is the one that guesses as high as possible without exceeding the true cost. Since h2 is just h1 with more restrictions and additional constraints, it's a more informed heuristic, and consistently gives estimates that are either equal to or higher than those of h1.

Compare how much of the search space each of the three approaches (brute-force, best-first with heuristic 1,
best-first with heuristic 2) explores in searching for a solution (number of nodes visited), for the puzzles provided.

Use tables, figures and graphs similar to the ones you made for Assignment 1.

Puzzle	**BFS**	**H1**	**H2**
Beginner (01)	1077	1067	1045
Intermediate (11)	860	854	851
Advanced (23)	2823	2408	2028
Expert (31)	3999	3946	3840
Average (across all 40)	3238	2850.875	2610.675



As we can see from the graph above, and the table, BFS consistently explores the largest portion of the search space, since it expands states by depth without using any heuristics. It has the highest average number of nodes visited. Both H1 and H2 visit lesser nodes overall, which shows that heuristic best-first search reduces the number of nodes explored by a big amount when compared to brute-force BFS. We can also see that between H1 and H2, H2 visits lesser nodes, since it has additional information about blocking vehicles and so narrows the search even further.

Include a disclosure in your report that describes the ways and extent to which you used AI.

Mainly for debugging and also a little bit to understand the assignment specifications and to better understand A* search.

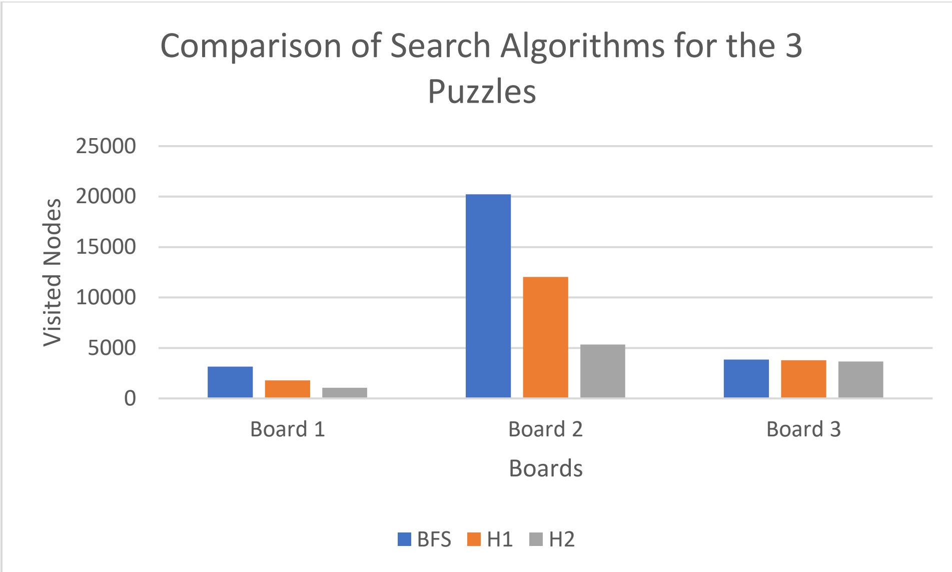
PART B

Describe how you designed the puzzles to make them difficult.

I started with just the red car on the board, and added cars to block it from reaching the exit. For every car I added, I added another car to block that car. This increased the number of dependent moves required to find a solution and the number of intermediate states that needed to be explored, especially for breadth-first search, which has to expand all states at smaller depths before reaching the solution.

As we can see from the clustered bar chart below, there is a much bigger difference in the number of nodes visited between BFS, H1, and H2 in the 2^{nd} Board. This is because of the many layered blocking dependencies which cause a large increase in solution depth. BFS performed the worst (as expected), and H2 performed the best because it accounts for blocking vehicles specifically.

Measure their difficulty (report the number of nodes visited for the three algorithms).



Number of Nodes Visited For Each Algorithm Per Board

	Board 1	**Board 2**	**Board 3**
BFS	3167	20215	3852
H1	1777	12032	3802
H2	1058	5349	3655